

Xiang LIU

Homepage: <https://xiangli-ml.github.io/>

Google Scholar: [VtK5lwUAAAAJ](#) (1106 citations)

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EDUCATION

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- **The Hong Kong University of Science and Technology (Guangzhou)** **Guangzhou, CHINA**
PhD in DSA Thrust, supervised by Professor Xiaowen Chu *Sep 2023 - Aug 2027*
 - **New York University** **NY, USA**
Visiting PhD in CDS, supervised by Professor Eunsol Choi *July 2025 - Jan 2026*
 - **The University of Hong Kong** **Hong Kong SAR, CHINA**
Master of Science(MSc) in Computer Science *Sep 2022 - Aug 2023*
 - **George Mason University** **VA, USA**
Bachelor of Science in Computer Science; GPA: 3.71/4.0 *Aug 2018 - Aug 2022*
Honors/Awards: Dean's List (2018-2020)

SELECTED RESEARCH

First / co-first author: 10 papers (ICML $\times 2$ · ICLR $\times 2$ · NeurIPS $\times 3$ · EMNLP Findings $\times 2$ · Workshop $\times 1$)

- **SIM: Benchmarking and Preserving High-Density Reasoning in KV Cache Compression** **ICML 2026**
First Author · KV Cache, Reasoning, Compression, Benchmark
- **MDN: Parallelizing Stepwise Momentum for Delta Linear Attention** **ICML 2026**
Co-first Author · Linear Attention, Delta Rule, Parallelization, Momentum
- **DiffAdapt: Difficulty-Adaptive Reasoning for Token-Efficient LLM Inference** **ICLR 2026**
First Author · Adaptive Inference, Reasoning LLM, Token Efficiency, Entropy Analysis
- **Reasoning Language Model Inference Serving Unveiled: An Empirical Study** **ICLR 2026**
Co-first Author · LLM Serving, Reasoning System, Empirical Study, Memory Optimization
- **ChunkKV: Semantic-preserving KV cache compression for efficient long-context LLM** **NeurIPS 2025**
First Author · KV Cache, Inference, GPU Memory, Long-Context
- **Perovskite-LLM: Knowledge-Enhanced LLMs for Solar Cell Research** **EMNLP Findings 2025**
Co-first Author · Domain-Specific LLM, Knowledge Augmentation, Materials Science
- **LISA: Layerwise Importance Sampling for Memory-Efficient LLM Fine-Tuning** **NeurIPS 2024**
Co-first Author · Parameter Efficient Fine-tuning, Layer-wise Optimization
- **Should We Really Edit Language Models? On the Evaluation of Edited LLMs** **NeurIPS 2024**
Co-first Author · Model Editing, Benchmark, Model Robustness
- **LongGenBench: Long-context Generation Benchmark** **EMNLP Findings 2024**
First Author · Long-context Generation, Long-context LLMs, Logical Coherence

EXPERIENCE

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- **Mind Lab** **Remote**
Research Intern *April 2026 - Present*
 - Post-trained **GLM-5.1** and **Qwen3.6-35B** for tool use and code generation, lifting agentic performance on VITABENCH, PINCHBENCH, TAU3BENCH, SWE-BENCH, and TERMINAL2BENCH.
 - Strengthened the model harness for long-horizon, multi-turn tool calling and code execution, improving robustness over extended trajectories.
 - Co-authored the δ -mem, MinT, and Macaron v1 technical reports.
 - **NYU Center for Data Science** **NY, USA**
Visiting Student Supervisor: Prof. Eunsol Choi *July 2025 - Jan 2026*
 - Proposed **DiffAdapt**, a novel algorithm for efficient adaptive reasoning for LLMs (Accepted to ICLR 2026).
 - Conducted comprehensive entropy analysis on reasoning traces to identify "overthinking" phenomena.
 - **HKUST Statistics and Machine Learning Research Group** **HK, CHINA**
Research Intern Supervisor: Prof. Tong Zhang *Dec 2022 - Aug 2023*
 - Proposed **LISA**, a novel algorithm for efficient fine-tuning of LLMs, accepted at *NeurIPS 2024*.
 - Contributed to the LMFlow framework, enabling personalized LLM fine-tuning and deployment (LLaMA, Bloom, Vicuna).
 - Conducted research on Chain-of-Thought (CoT) methods to enhance logical reasoning in LLMs.
 - **Baidu Research Cognitive Computing Lab** **Beijing, CHINA**
Research Intern *Dec 2021 - June 2022*
 - Worked on dependency parsing using Open Information Annotation (OIA) to convert sentences into DAGs.

COMPETITIONS

- **Kaggle — Feedback Prize - Predicting Effective Arguments Competition** **Jun 2022 - Aug 2022**
 - *Team Leader — Silver Medal (Top 2%)*
 - Developed baseline code, designed data preprocessing strategies, and optimized model structures.
 - Implemented token classification replacing sequence classification, boosting rank and efficiency.

PROFESSIONAL SKILLS

- **Languages:** English, Mandarin
- **Computing Skills:** PyTorch, Git, Linux